

Tutorial Notes on Maximum Likelihood Estimation (MLE)

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1. Random Variable

A **random variable** is a variable whose value depends on the outcome of a random experiment.

Let X be a continuous random variable. We observe a set of n samples:

$$x_1, x_2, \dots, x_n$$

These observations are assumed to be generated by an unknown probabilistic process.

2. Probability Distribution

A **probability distribution** describes how the values of a random variable are distributed.

In Maximum Likelihood Estimation, we assume a known distributional form with unknown parameters.

In this tutorial, we assume:

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

where:

- μ is the mean (unknown)
- σ^2 is the variance (unknown)

3. Probability Density Function

The **probability density function (PDF)** of a Gaussian random variable is given by:

$$p(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

4. Joint Probability

Suppose we observe n independent and identically distributed (i.i.d.) samples:

$$x_1, x_2, \dots, x_n$$

The joint probability density function is:

$$p(x_1, x_2, \dots, x_n \mid \mu, \sigma^2) = \prod_{i=1}^n p(x_i \mid \mu, \sigma^2)$$

5. Likelihood Function

The **likelihood function** treats the observed data as fixed and the parameters as variables.

$$L(\mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

This can be rewritten as:

$$L(\mu, \sigma^2) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right)$$

6. Log-Likelihood Function

To simplify the computation, we take the natural logarithm of the likelihood function.

$$\begin{aligned} \ell(\mu, \sigma^2) &= \log L(\mu, \sigma^2) \\ &= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \end{aligned}$$

The constant term $-\frac{n}{2} \log(2\pi)$ does not affect maximization.

7. Maximization of the Function

To obtain the Maximum Likelihood Estimates, we differentiate the log-likelihood function and set the derivatives equal to zero.

7.1 Derivative with respect to Mean μ

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu)$$

Setting the derivative to zero:

$$\sum_{i=1}^n (x_i - \mu) = 0$$

$$\boxed{\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i}$$

7.2 Derivative with respect to Variance σ^2

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n (x_i - \mu)^2$$

Setting the derivative to zero:

$$-n\sigma^2 + \sum_{i=1}^n (x_i - \mu)^2 = 0$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})^2$$

8. Maximum Likelihood Estimation

The parameter values that maximize the likelihood function are called the **Maximum Likelihood Estimates (MLEs)**.

$$\hat{\mu}_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^n x_i \quad \hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})^2$$

Illustrative Example

Consider the dataset:

10, 12, 14, 16, 18

Step 1: Mean Estimation

$$\hat{\mu} = \frac{10 + 12 + 14 + 16 + 18}{5} = 14$$

Step 2: Variance Estimation

$$\begin{aligned} \hat{\sigma}^2 &= \frac{(10 - 14)^2 + (12 - 14)^2 + (14 - 14)^2 + (16 - 14)^2 + (18 - 14)^2}{5} \\ &= \frac{16 + 4 + 0 + 4 + 16}{5} = 8 \end{aligned}$$

Final Result

$$\mu = 14 \quad \sigma^2 = 8$$

Important Notes

- MLE maximizes likelihood, not unbiasedness.
- Variance MLE uses $\frac{1}{n}$ instead of $\frac{1}{n-1}$.

- Log-likelihood simplifies differentiation.
- MLE is consistent and efficient for large samples.